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ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced)-2016

ANSWERS, HINTS & SOLUTIONS
HALF COURSE TEST–VII
(Paper - 1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**B**	**A**	**C**
2.	**B**	**A**	**C**
3.	**A**	**A**	**A**
4.	**C**	**A**	**B**
5.	**B**	**A**	**A**
6.	**A**	**D**	**C**
7.	**A**	**D**	**A**
8.	**A, C**	**A, B, D**	**A, B, C**
9.	**A, C**	**A, B, C, D**	**A, C**
10.	**B, C, D**	**A, B, C, D**	**A, B, C, D**
11.	**A, B, C, D**	**A, B, C, D**	**A, B**
12.	**B**	**C**	**C**
13.	**A**	**A**	**A**
14.	**C**	**A**	**C**
15.	**D**	**B**	**B**
16.	**A**	**A**	**D**
1.	**(A → p); (B → q); (C → s); (D → p)**	**(A → t); (B → s); (C → p); (D → q, r)**	**(A → q); (B → r); (C → p) (D → s)**
2.	**(A → s); (B → p); (C → q); (D → r)**	**(A → q, r); (B → p); (C → s); (D → t)**	**(A → s) (B → p); (C → q); (D → r)**
1.	**7**	**7**	**4**
2.	**1**	**2**	**6**
3.	**3**	**1**	**1**
4.	**4**	**4**	**3**
5.	**2**	**4**	**0**
6.	**6**	**2**	**1**

Physics

PART – I

Hints & Solution

SECTION – A

1. Pressure in the air inside the column of mercury is equal to the weight of mercury over the air divided by the internal cross sectional area of the tube. When the temperature increases, the weight of the upper part of the mercury column does not change. That is why the pressure in the air is also constant. For the isobaric process, the change in volume is proportional to the change in temperature. The same is true for the lengths of the air column.

$$\frac{l}{l_0}=\frac{T}{T_0} \quad \Rightarrow \quad l=\frac{l_0 T}{T_0}=11 \text{ cm}.$$

2. Frequency received by car B is $f_1=\left(\frac{v+w+u}{v+w-u}\right)f$

Now the car B will be treated as a source of frequency f_1

$\therefore$ wavelength of reflected sound received by the driver of car A is

$$\lambda'=\frac{(v-w-u)}{f_1}=\frac{(v-w-u)(v+w-u)}{(v+w+u)f}$$

$$\frac{(330-5-25)(330+5-25)}{(330+5+25)300}=\frac{31}{36}m$$

Hence (B) is correct

3. $$a=\frac{F_0}{m}e^{-Kx} \Rightarrow \int_0^{v_{max}} v\,dv=\frac{F_0}{m}\int_0^{\infty}e^{-Kx}dx$$

$$\Rightarrow \quad \frac{v_{max^2}}{2}=\frac{F_0}{mK} \Rightarrow K_{max}=\frac{F_0}{K}$$

4. Let initial location of center of mass is 0 then find location of cm

$$X_{cm}=\frac{\lambda(90\,cm)(5\,cm)+\lambda(10\,cm)(45)}{\lambda(90)+\lambda(10)}=9 \text{ cm}$$

5. As escape velocity is $\sqrt{2}$ times of orbital velocity.

6. The power developed by the man = power required in the reference frame of the man
= Force $\times$ velocity = m(g + a).(h/t)
Hence, (A) is correct.

7. $x + y = 100$ …(1)
Heat lost = Heat gained
$y \times 3.36 \times 10^5 = x \times 21 \times 10^5$

$$\frac{y}{x}=\frac{21}{3.36}$$

$336y = 21x \times 100$ …(2)
Using (1) & (2)

$$y=\frac{210000}{2436}=86.21$$

8. For a plane wave intensity (energy crossing per unit area per unit time) is constant at all point
For spherical wave

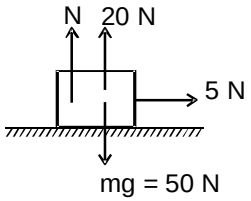
$I = \frac{p}{4\pi r^2} \Rightarrow \left(I \propto \frac{1}{r^2}\right)$

9. Acceleration of block is = 10 m/s^2

$\Rightarrow$ Displacement $s = \frac{1}{2}at^2 = \frac{1}{2} \times 10 \times \frac{4}{10} = 2m$

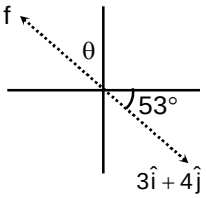
Tension in the string is 40 N

10. N = 50 – 20 = 30 N
Limiting friction force = μN = 12 N and applied force in horizontal direction is less than the limiting friction force, therefore the block will not slide.
For equilibrium in horizontal direction, friction force must be equal to 5 N.



From the top view, it is clear that $\theta = 37°$ i.e. 127° from the x-axis that is the direction of the friction force. It is opposite to the applied force.

Contact force $= \sqrt{N^2 + f^2} = \sqrt{925}\,N$



11. $P = 3V^3$, for ideal gas
$PV = nRT$
(A) relation between V and T

$V \times (3V^3) = nRT \Rightarrow V^4 = \left(\frac{nR}{3}\right)T \qquad V^4 \propto T$

(B) Relation between P and T
$PV = nRT$

$\Rightarrow \; P\left(\frac{P}{3}\right)^{1/3} = nRT \Rightarrow P^{4/3} \propto T$

(C) For expansion
$V \rightarrow$ increases
work-done : positive
internal energy : increases
Hence, heat will have to supplied to the gas.

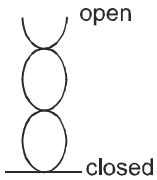
(D) As $T \propto V^4$
with increase in temperature, volume increases
Hence work done is positive.

12. $\frac{2\pi}{0.8} = \frac{2\pi}{\lambda}$ So $\lambda = 0.8$ and effective length of air column = 0.99 + 0.01 = 1m.

$\frac{\ell}{\lambda} = \frac{5}{4}$

$l = 5\,\frac{\lambda}{4}$, so five half loops will be formed

$\ell = 5\left(\frac{\lambda}{4}\right)$ so second overtone



13. $P_{ex} = -B\,\frac{d\xi}{dx}$

$= (5 \times 10^5) \times (0.1 \times 10^{-3})\,\frac{2\pi}{0.8}\,\sin\,\frac{2\pi}{0.8}\,(y + 1cm)\cos 2\pi(400)\,t$

$= (125\,\pi\ N/m^2)\sin\,\frac{2\pi}{0.8}\,(y + 1cm)\cos 2\pi(400t)$

14- 16. $\frac{m}{4}.\frac{2\ell}{3}.v_0 = \frac{m\ell^2}{3}\omega + \frac{m}{4}v_1\frac{2\ell}{3}$

$2v_0 = \frac{8\omega\ell}{3} \Rightarrow \omega = \frac{3v_0}{4\ell}$

$v_1 = \frac{2\omega\ell}{3} - v_0$

$v_1 = \frac{-v_0}{2}$

$\frac{mg\ell}{2}(1-\cos\theta) = \frac{1}{2}I\omega^2$

SECTION – C

1. $P = \left(\frac{0.16\times10^{-6}}{60}\right)\times1000\times4.2\times80\times10^3 + \left(\frac{0.16\times10^{-6}}{60}\right)\times1000\times2.3\times10^{+6} = 7$ watt.

2. $T\left(\frac{h}{2}\right) - N\left(\frac{h}{4}\right) > 0$

$T - \frac{N}{2} > 0$

$T_{min} = \frac{Mg}{2}$

$a = g/2 = 5\ m/s^2.$

$mg - T = m(g/2)$

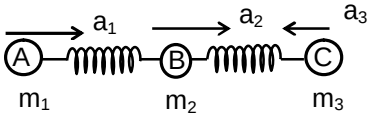
$mg - \frac{Mg}{2} = m\left(\frac{g}{2}\right)$

$\frac{Mg}{2} = \left(\frac{mg}{2}\right)$

$\frac{m}{M} = 1$

3. The acceleration of center of mass of the system

$a_{cm} = \frac{|m_1\vec{a}_1 + m_2\vec{a}_2 + m_3\vec{a}_3|}{m_1 + m_2 + m_3}$



$\Rightarrow$ The net force acting on the system

$= |m_1\vec{a}_1 + \vec{m}_2a_2 + m_3\vec{a}_3|$

$\Rightarrow F_{net} = (m_1 a_1 + m_2 a_2 - m_3 a_3)$

$= \left[(1)(1) + (2)(2) - \frac{1}{2}(4)\right]N = 3\ N.$

4. $R_{max} = \frac{u^2}{g}$

So, maximum area covered = $\pi R_{max}^2 = \pi . \frac{u^4}{g^2} = 4m^2$

5. Velocity of block w.r.t. ground

$\vec{V}_{1G} = 20\hat{i} + 15\hat{j}$

$\Rightarrow \vec{V}_{1P} = 15\hat{j}$

Velocity of block w.r.t. ground after 10 sec. is zero (since block will stop w.r.t. board after 5 sec.)

$\vec{V}'_{1P} = \vec{V}_{1P} - \mu g t \hat{j} = 0$ (t = 5 sec)

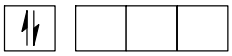
$\therefore \vec{V}'_{1G} = 20\hat{i}$

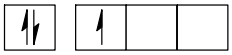
6. Area under graph = $\int \frac{1}{v} dx = \Delta t$

Chemistry

PART – II

1. $E = \dfrac{\text{Molecular weight}}{\text{Replaced } H^+} = 98/1 = 98$

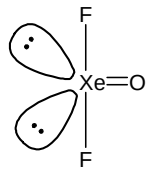
2. B^+ : [He] 

   C^+ : [He] 

   Sr^+, Ba^{2+} are relatively easily formed in order to achieve completely filled valence shell orbitals.

3. $K_c^{-1} = \dfrac{\left[Ag^+\right]\left[NH_3\right]^2}{\left[Ag\left(NH_3\right)_2^+\right]}$

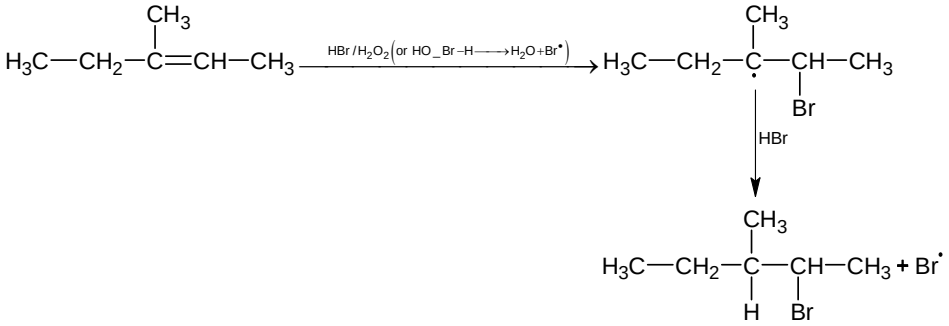
   $K_{sp} = \left[Ag^+\right]\left[Cl^-\right] \Rightarrow \dfrac{1}{K_c} = \dfrac{K_{sp}\left[NH_3\right]^2}{\left[Ag\left(NH_3\right)_2^+\right]\left[Cl^-\right]}$

4. $XeOF_2$: 

   Hence,

   ⊢ or ⊤

5. $\lambda = \dfrac{h}{mv} = \dfrac{6.626\times10^{-34}}{\left(9.1\times10^{-31}\right)\left(2.19\times10^{6}\right)} = 3.321\times10^{-10}\ \text{m}$

6. 

7. $CO(g) + \dfrac{1}{2}O_2(g) \longrightarrow CO_2(g)$

   $\Delta G^0 = \Delta G^0\left(CO_2\right) - \left[G^0(CO) + \dfrac{1}{2}G^0\left(O_2\right)\right] = -394.4 - (-137.2 + 0) = -257.2\ \text{kJ}$

   $\Rightarrow$ Spontaneous (–ve sign)

   But $\Delta G^0 = \Delta H^0 - T\Delta S^0$

   $\Rightarrow -257.2 = \Delta H^0 - 298\ K(-0.094)$

$\Rightarrow \Delta H^0 = -285.2$ kJ $\Rightarrow$ exothermic (–ve sign)

8. Allyl carbocation is less stable than benzyl carbocation due to less number of resonating structures.

9. In (A) Cl and O
In (B) S and O
In (C) only O
In (D) P and O

10. (A)  (B)  (C)  (D) CHI_3 (E) CH_3COO^- (E) CH_3COOH

11. The balanced equation is

$$\underset{294\text{ g}}{K_2Cr_2O_7} + \underset{14\times36.5\text{ g}}{14HCl} \longrightarrow 2KCl + 2CrCl_3 + 7H_2O + \underset{3\times71\text{g}}{3Cl_2}$$

Mass of $K_2Cr_2O_7 = \frac{96}{100} \times 61.3 = 58.848$g

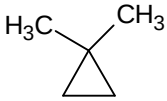
Mass of HCl = $\frac{30}{100} \times 320 \times 1.15 = 110.4$g

Hence, $K_2Cr_2O_7$ is limiting reagent.
$\therefore$ Mass of Cl_2 produced = 42.63 g

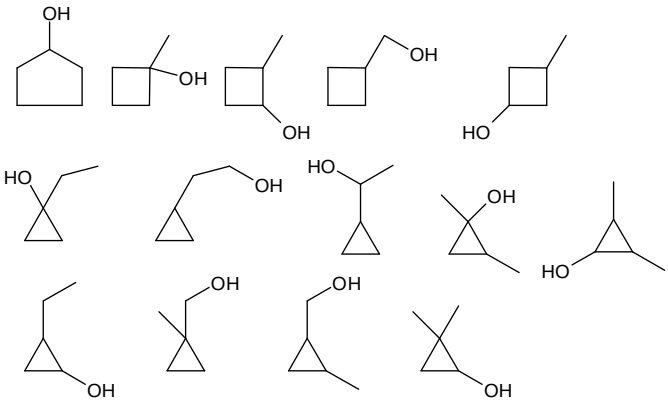
Also equivalent of $K_2Cr_2O_7 = \frac{M}{6}$

Equivalent of HCl = $\frac{7M}{3}$

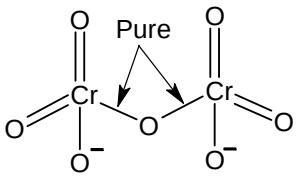
16. All the given molecules are cyclic isomers with molecular formula = C_5H_{10}
Number of pairs of diastereomers
= total number of pairs of stereomerms – number of pairs of enantiomers = $^3C_2 - 1 = 2$

 is th missing cyclic isomer.

SECTION – C

1. 

x = 14 so x/2 = 7

2. 

As these two does not participate in resonance.

3. It will produce most stable free radical.

4. meq of KOH = 20 meq = meq of acid

$$\frac{20}{1000} = \frac{0.45}{90} \times n - \text{factor}$$

n-factor for acid = 4 = number of replaceable proton.

5. $[H_2CO_3]$ in blood = 4M $[NaHCO_3]$ = 10 M
Volume of blood = 10 mL
Let the volume of $NaHCO_3$ used = x mL

$$\left[H_2CO_3\right] \text{ in mixture} = \frac{4 \times 10}{(x+10)}$$

$$\left[NaHCO_3\right] \text{ in mixture} = \frac{10 \times x}{(x+10)}$$

$$pH = pK_a + \log\frac{\text{salt}}{\text{acid}}$$

$$7.4 = 7 - \log 4 + \log\frac{(10 \times x)(x+10)}{(x+10)(4 \times 10)} \Rightarrow 7.4 = 6.4 + \log\frac{10x}{40}$$

$$\log\frac{10x}{40} = 1 \text{ so } \frac{10x}{40} = 10$$

x = 40 mL
Hence x/10 = 4

6. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

1 3
0 0 2

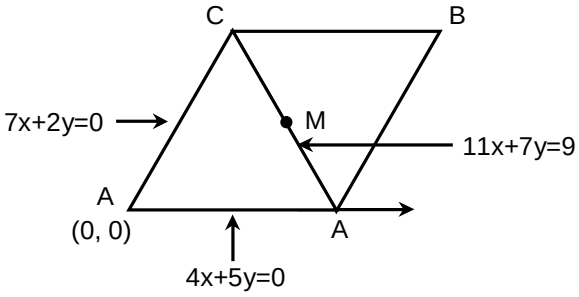
$\therefore$ Ratio of initial and final volumes = 4/2 = 2

Mathematics

PART – III

SECTION – A

1. Since O(0, 0) does not lie on 11x + 7y = 9
hence it is the equation of AC,



solving OC with AC we get $C \equiv \left(-\frac{2}{3},\frac{7}{3}\right)$

solving OA with AC we get $A \pm \left(\frac{5}{3},-\frac{4}{3}\right)$

hence M = mid point of C & A $\equiv \left(\frac{1}{2},\frac{1}{2}\right)$

hence equation of OB $\equiv$ y = x.

2. The curve remains unchanged when y & x are interchanged.

3. Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$, if $\left(m_i,\frac{1}{m_i}\right)$ lies on it then

$m^4 + 2gm^3 + 2fm + cm^2 + 1 = 0$ and $m_1\, m_2, m_3, m_4$ are its roots.

4. Coefficient of x^r is $\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)...\left(-\frac{1}{2}-r+1\right)}{r!}(-2)^r$

$$= \frac{1.3.5...(2r-1)}{2^r}.\frac{(-1)^r(-1)^r 2^r}{r!} = \frac{1.3.5...(2r-1)}{r!} = \frac{(2r)!}{r!r!}2^r$$

5. $\frac{1}{2}+\frac{3}{8}+\frac{9}{32}+\frac{27}{128}+...\infty = \frac{\frac{1}{2}}{1-\frac{3}{4}} = 2.$

6. $\frac{1-a^{x+1}}{1-a} = (1+a)\left(1+a^2\right)\left(1+a^4\right)$

i.e. $1-a^{x+1} = 1-x^8$
(multiply by (1 – a) on both the sides)
$\therefore$ x = 7.

7. ABC can speak in ${}^{10}C_3 \times 1$ ways
other 7 in 7! ways
so totally ${}^{10}C_3$. 7! ways $= \frac{10!}{6}$.

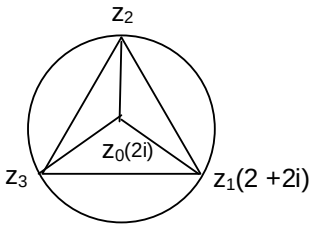
8. Putting $x = \omega$ in the equation,
$0 = a_0 + a_1\,\omega + a_2\omega^2 + a_3 + \ldots$ .... (1)
Putting $x = \omega^2$ in the equation,
$0 = a_0 + a_1\omega^2 + a_2\omega + a_3 + \ldots$ .... (2)
Putting x = 1 in the equation,
$3^n = a_0 + a_1 + a_2 + a_3 + \ldots$ .... (3)
adding (1), (2) and (3),
$3^n = 3(a_0 + a_3 + a_6 + \ldots.)$ .... (a)
$\Rightarrow a_0 + a_3 + a_6 + \ldots. = 3^{n-1}$ (option C)
subtracting (2) from (1),
$0 = (\omega - \omega^2)(a_1 - a_2 + a_4 - a_5 + \ldots)$
Since $\omega - \omega^2 \neq 0$, $a_1 + a_4 + a_7 + \ldots = a_2 + a_5 + a_8 + \ldots$ ... (4)

Also from (3) – (a), $a_1 + a_2 + a_4 + a_5 + \ldots = 3^n - 3^{n-1} = 2.3^{n-1}$ ....(5)
From (4) and (5), $a_1 + a_4 + a_7 + \ldots = a_2 + a_5 + a_8 + \ldots = 3^{n-1} = a_0 + a_3 + a_6 + \ldots$

9. Clearly the inscried triangle is equilateral.

$\Rightarrow \dfrac{z_2 - z_0}{z_1 - z_0} = e^{i\frac{2\pi}{3}}, \dfrac{z_3 - z_0}{z_1 - z_0} = e^{-i\frac{2\pi}{3}}$

$\Rightarrow z_2 = -1 + i(2 + \sqrt{3})$ and $z_3 = -1 + i(2 - \sqrt{3})$



10. b is the H.M. of a and c and their G.M. = $\sqrt{ac}$

G.M > H.M $\Rightarrow \sqrt{ac} > b$

And A.M. of a^n and c^n > G.M. of a^n and c^n

$$\frac{a^n + c^n}{2} > \sqrt{a^n c^n}$$

$$a^n + c^n > 2\left(\sqrt{ac}\right)^n > 2b^n$$

Put n = 100, 3, 5, 2

11. $\sqrt{1 - \sin A} = \left|\sin\dfrac{A}{2} - \cos\dfrac{A}{2}\right| = \sin\dfrac{A}{2} - \cos\dfrac{A}{2}$ if $\sin\dfrac{A}{2} \geq \cos\dfrac{A}{2}$

ie $\dfrac{\pi}{4} \leq \dfrac{A}{2} \leq \dfrac{5\pi}{4} \Rightarrow 0 \leq \dfrac{A}{2} - \dfrac{\pi}{2} \leq \pi$.

12. 6 can be written in the following ways using 1's and 2's only.

Six 1's in 1 way

Four 1's and one 2's in $\dfrac{5!}{4!}$ = 5 ways

Two 1's and two 2's in $\dfrac{4!}{2!\,2!}$ = 6 ways

Three 2's in 1 way.

So f(6) = 1 + 5 + 6 + 1 = 13.

13. We have f(6) = 13, so f{f(6)} = f(13).

Now 13 can be expressed in the following ways:

Thirteen 1's in 1 way

Eleven 1's and one 2's in $\dfrac{12!}{11!}$ = 12 ways

Nine 1's and two 2's in $\dfrac{11!}{9!\,2!}$ = 55 ways

Seven 1's and three 2's in $\dfrac{10!}{7!\,3!}$ = 120 ways

Five 1's and four 2's in $\dfrac{9!}{5!\,4!}$ = 126 ways

Three 1's and five 2's in $\dfrac{8!}{3!\,5!}$ = 56 ways

One 1's and six 2's in $\dfrac{7!}{6!}$ = 7 ways.

$\therefore$ f(13) = 377.

14–16. $\lambda x - y + 2(1 + \lambda) = 0 \Rightarrow \lambda(x + 2) - (y - 2) = 0$ pass through (– 2, 2)

$\mu x - y + 2(1 - \mu) = 0 \Rightarrow \mu(x - 2) - (y - 2) = 0$ pass through (2, 2).

Clearly these represent the foci of ellipse, so 2ae = 4.

The circle $x^2 + y^2 - 4y - 5 = 0 \Rightarrow x^2 + (y-2)^2 = 9$ represents auxiliary circle thus $a^2 = 9 \Rightarrow e = \frac{2}{3}$ and $b^2 = 5$.

SECTION – B

1 (A). $|a + ib|\,|z| = |z|\,|(a-1) + ib|$

$\Rightarrow \frac{1}{\sqrt{2}} = \sqrt{(a-1)^2 + b^2}$ and $a^2 + b^2 = \frac{1}{2}$

$\Rightarrow 1 - 2a = 0 \Rightarrow a = \frac{1}{2}$

and $b^2 = \frac{1}{4} \Rightarrow b = \frac{1}{2}$

$\Rightarrow a - b = 0$.

(B) We must have either
$6a > 2^a \cdot 3^b$ or $6b > 2^a \cdot 3^b$.
If b = 0, then $6a > 2^a \Rightarrow a = 1, 2, 3, 4$.
If a = 0, then $6b > 3^b \Rightarrow b = 1, 2$
suppose a > 0 and b > 0 $\Rightarrow$ if $6a > 2^a \cdot 3^b$
thus $6^a > 2^a \cdot 3$ so $2a > 2^a$ (not possible)
similarly if $6b > 2^a \cdot 3^b$ than $3b > 3^b$ not possible
$\Rightarrow$ only solutions are 2, 4, 8, 76, 3, 9.

(C). $\mathrm{Re}\left(\frac{z}{4}\right) \in (0, 1)$, $\mathrm{Im}\left(\frac{z}{4}\right) \in [0, 1)$

means that if z = a + ib than a, b $\in$ (0, 4)

Now $\frac{4}{a - ib} = \frac{4a}{a^2 + b^2} + \frac{4bi}{a^2 + b^2}$

$\Rightarrow 0 < a, b < \frac{a^2 + b^2}{4}$

$\Rightarrow (a-2)^2 + b^2 > 4$ and $a^2 + (b-2)^2 > 4$
so we want area inside the square and outside the two circles
$\Rightarrow$ area $= 16 - 4\pi + (2\pi - 4)$
$12 - 2\pi$.

(D). $3x + 4y + z = 5$

let $\vec{a} = 3\hat{i} + 4\hat{j} + \hat{k}$

$\vec{b} = x\hat{i} + y\hat{j} + z\hat{k}$

$\Rightarrow (\vec{a} \cdot \vec{b}) \le |\vec{a}||\vec{b}|$

$\Rightarrow 5 \le \sqrt{9 + 16 + 1} \cdot \sqrt{x^2 + y^2 + z^2}$

$\Rightarrow 26\left(x^2 + y^2 + z^2\right) \ge \frac{25}{26}$

2 (A) $\sqrt{\sin(\cos x)} = \sin(\cos x) \ge 0$

$2n\pi \le \cos x \le (2n + 1)\pi$

$0 \le \cos x \le 1$

$x \in \left[2n\pi - \frac{\pi}{2},\ 2n\pi + \frac{\pi}{2}\right]$.

(B) $\left(\sqrt{\cos(\sin x)}\right)^{-1}$

cos(sinx) > 0

$-\frac{\pi}{2} < \sin x < \frac{\pi}{2} \Rightarrow x \in R.$

(C) $\tan(\pi \sin x)$

$\pi \sin x \neq \pm\frac{\pi}{2}$

$\sin x \neq \pm\frac{1}{2}$

$\therefore x \neq n\pi \pm \frac{\pi}{6}$

$x \in R - \left\{n\pi \pm \frac{\pi}{6}\right\}.$

(D) $\ln(\tan x)$

$\tan x > 0$

$0 < x < \frac{\pi}{2}$

$n\pi < x < n\pi - \frac{\pi}{2}.$

SECTION – C

1. $x = \frac{1/4}{1-\frac{1}{2}} = \frac{1}{2} \therefore \left(\frac{1}{5}\right)^{\log\sqrt{5}\left(\frac{1}{2}\right)} = 5^{-\log 5^{(4^{-1})}} = 4$

2. $\alpha + \beta = -\frac{7}{2},\ \alpha\beta = \frac{c}{2} \Rightarrow \alpha^2 - \beta^2 = \pm\frac{7}{4}$

i.e. $(\alpha+\beta)(\alpha-\beta) = \pm\frac{7}{4}$ i.e. $-\frac{7}{2}\sqrt{\frac{49}{4} - 2x} = \pm\frac{7}{4} \Rightarrow c = 6.$

4. The given expression can be written as

$= k^3\left(\sum \sin^3 A \cos(B-C)\right) = \frac{k^3}{2}\sum \sin^2 A(\sin 2B + \sin 2C)$

$= k^3 \sum \sin A \sin B(\sin A \cos B + \cos A \sin B)$

$= k^3 \sum \sin A \sin B \sin C = k^3 3 \sin A \sin B \sin C = 3(abc).$

5. Line 'L' passes through the centre of second circle.

6. $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0 \Rightarrow a(b-1)(c-1) - (b-1)(1-a) - (c-1)(1-a) = 0$

$\Rightarrow \quad \frac{a}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 0.$